Group - III crossection derivation

Corrections (blue = new, red = remove; see Group_check.pdf):

- Row III, first line ($\mathfrak{A}^\dagger\bar{\mathfrak{C}}^\dagger\mathfrak{B}$ form): the overall sign is $-$, as you already have in Rows V and VI (Rows III, V, VI enter with $-\mathfrak{C}'^\dagger$). The explicit forms below it are right.
- Row III, second part (C5): the measure is $\int_{w,w',x',y',z,x}$ before and $\int_{w,x',y',z,x}$ after the w' integral (y does not appear in this part; x' does).
- Shown in blue after Eqs. (1.7), (1.10), (1.14), (1.17) (nothing changes): the BFKL term (B) $\propto \log\frac{\vee}{k^+}$ of Rows III (both parts), V, VI, which sits inside line 1, and each row as (B) + a part that is finite for $\vee \to \infty$.

1 Group III (with c.c.)

1.1 Row I [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\left[\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)U^{c'b}(z)\right.$$
$$\left. - \mathbb{A}_i^{\dagger\,(1)\,c'}(k^+,-w')U^{bd'}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,d'}(p^+,-z)U^{c'a}(w')\right]\times\left[U^{ac}(w)\mathbb{B}_{2ji}^{bc}(k^+,-w;p^+,-z) - U^{\dagger\,bc}(z)\bar{\mathbb{B}}_{2ji}^{ca}(k^+,-w;p^+,-z)\right]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\frac{1}{[p^+(x-z)^2+k^+(x-w)^2]}$$
$$\left[\frac{\delta^{ij}\left[(x-z)^2-(x-w)^2\right]}{2(p^++k^+)(z-w)^2}+\frac{1}{p^+}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j(z-w)^i}{(z-w)^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j(x-w)^i}{2(x-w)^2}\right]\right]\times\left[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\right] \tag{1.1}$$

Integrating w.r.t. p^+ and simplifying we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x-w)^2+\Lambda(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right.$$
$$\left.-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\vee}{k^++\Lambda}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$$
$$-\frac{ig^4}{8\pi^5}\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\left.\right\}\times$$
$$\left[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)\right]\times$$
$$\left[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\right] \tag{1.2}$$

1.2 Row II [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\left[\bar{\mathbb{A}}_j^{\dagger\,(1)\,c'}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')U^{bc'}(z) - \mathbb{A}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')\right]\times$$
$$\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z) - U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\frac{1}{\left[p^+\,(x-z)^2+k^+\,(x-w)^2\right]}$$
$$\left[\frac{\delta^{ij}\,\left[(x-z)^2-(x-w)^2\right]}{2\,(p^++k^+)\,(z-w)^2}+\frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}\right.\right.$$
$$\left.\left.-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right]\times\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.3}$$

Integrating w.r.t. p^+ and simplifying we get

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x-w)^2+\Lambda\,(x-z)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.$$
$$\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]$$
$$+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{k^++\Lambda}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}$$
$$-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times$$
$$\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x') - U^{ab'}(x')\,\rho^b(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.4}$$

1.3 Row III [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{4}{(2\pi)^3}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\left[\mathbb{A}_{k'}^{\dagger\,(1)\,e'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)\right.$$
$$\left.-\bar{\mathbb{A}}_{k'}^{\dagger\,(1)\,c'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\right]\times\left[U^{ac}(w)\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)\right.$$
$$\left.-U^{\,bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}(k^+,-w;p^+,-z)\right]$$

We write it in two parts:

1.3.1 First part: [Done ✓]

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',x',y',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(w'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\delta^{(2)}\left[w'-\frac{(p^++k^+)\,y'-p^+\,z}{k^+}\right]\times\left[\delta_{k'm}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{jk'}\right]$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\}-U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{1.5}$$

Integrating w.r.t. w'

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{1.6}$$

Using + prescription, we get

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right) + \log\left(\frac{\vee}{k^+}-1\right)\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{1.7}$$

The BFKL term of this row, written out. It is inside line 1 of Eq. (1.7), in the $-\frac{1}{k^+}\,\delta_{im}\delta_{jk'}$ term: at large p^+ the phase $e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}$ lets only $y'\to z$ contribute to the y' integral, which gives $\frac{k^+}{p^++k^+}$ times the value at $y'=z$, and $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+} = \log\frac{\vee}{k^+}$ (`Why_log_vee.pdf`). Your equation stays as it is; it equals (B) + [your equation − (B)], with

$$\text{(B)} = -\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$

$$\text{Eq. (1.7)} - \text{(B)} = \text{Eq. (1.7)} + \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{(p^+ + k^+)}$$
$$\times\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$

(B) $\propto \log\frac{\vee}{k^+}$ is the BFKL term (it is in `BFKL_rows.pdf`); Eq. (1.7) − (B) has no $\log\frac{\vee}{k^+}$: it is finite for $\vee\to\infty$. The $\log(\frac{\vee}{k^+}-1)$ of your line 2 is not a BFKL term: it cancels the subtraction in $\left[\frac{1}{p^+}\right]_+$ of line 1.

1.3.2 Second part: [Done ✓]

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',x',y',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}}{(w'-z)^2\,(x'-y')^2}\,\frac{p^+}{(p^++k^+)\,[p^+\,(x-z)^2 + k^+\,(x-w)^2]}$$
$$\delta^{(2)}\left[\frac{p^+}{k^+}\,(y'-z) + y' - w'\right]\times\left[\frac{\delta_{k'm}\,d_\perp}{2\,(z-w)^2}\,[(x-z)^2-(x-w)^2] + \frac{(p^++k^+)\,\delta_{k'm}}{k^+}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}\right.\right.$$
$$-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2} - \frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\Big] + \frac{(p^++k^+)\,\delta_{k'm}}{p^+}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}\right.$$
$$\left.-\frac{[(x-z)^2-(x-w)^2]}{2\,(z-w)^2}\right] - \frac{(p^++k^+)^2}{p^{+\,2}}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{(p^++k^+)^2}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}\right.$$
$$\left.-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right] - \frac{(p^++k^+)^2}{p^+\,k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}\right.$$
$$\left.\left.+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.8}$$

Integrating w.r.t w'

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}$$
$$\Bigg[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.$$
$$\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]$$
$$-\frac{1}{p^+}\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]$$
$$-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\Bigg]\times$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.9}$$

Integrating w.r.t. p^+, using the plus prescription, we get

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2+k^+\,(x-w)^2]}$$
$$\Bigg[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^++k^+)^2}\,\left[(x-z)^2-(x-w)^2\right]+\frac{p^+\,\delta_{k'm}}{k^+\,(p^++k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2}-\frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.$$
$$\left.-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]+\frac{\delta_{k'm}}{(p^++k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2}+\frac{(x-z)\cdot(x-w)}{2\,(x-z)^2}-\frac{\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2}\right]$$
$$-\left[\frac{1}{p^+}\right]_+\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2}+\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right]-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2}-\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]$$
$$-\frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2}-\frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2}+\frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2}+\frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\Bigg]\times$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.$$
$$\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{1.10}$$

The BFKL term of this row, written out. It is inside line 1 of Eq. (1.10), in the $-\frac{p^+}{k^{+2}}$ term: at large p^+ the phase $e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}$ lets only $y'\to z$ contribute to the y' integral, which gives $\frac{k^+}{p^++k^+}$ times the value at $y'=z$, and $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\frac{\vee}{k^+}$ (`Why_log_vee.pdf`). Your equation stays as it is; it equals (B) + [your equation − (B)], with

$$\mathrm{(B)} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}$$
$$\times\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2}-\frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]$$
$$\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$

$$\text{Eq. (1.10)} - \text{(B)} = \text{Eq. (1.10)} + \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\,\frac{1}{(p^++k^+)}$$
$$\times\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}$$
$$\times\left[\frac{(y'-z)\cdot(z-w)}{(y'-z)^2\,(z-w)^2}-\frac{(y'-z)\cdot(x-w)}{2\,(y'-z)^2\,(x-w)^2}\right]$$
$$\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x')-f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$

(B) $\propto \log \frac{\vee}{k^+}$ is the BFKL term (it is in `BFKL_rows.pdf`); Eq. (1.10) – (B) has no $\log \frac{\vee}{k^+}$: it is finite for $\vee \to \infty$. The $\log(\frac{\vee}{k^+} - 1)$ of your line 2 is not a BFKL term: it cancels the subtraction in $\left[\frac{1}{p^+}\right]_+$ of line 1.

1.4 Row IV [Done ✓]

Only 4 ρ's

1.5 Row V [Done ✓]

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &-\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^+ + k^+, -y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y' - w'; p^+, w' - z)\\ &- \bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^+ + k^+, -y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^+ + k^+, y' - w'; p^+, w' - z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+, -z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+, -w)\\ &- U^{ac}(w)\bar{\mathbb{A}}^{(1)\,d}_{j}(p^+, -z)U^{db}(z)\mathbb{A}^{(1)\,c}_{i}(k^+, -w)\Big]\end{aligned}$$

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^{+\,2}}\int_{\Lambda}^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(w'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ &\delta^{(2)}\left[w' - \frac{(p^+ + k^+)\,y' - p^+\,z}{k^+}\right]\times\left[\delta_{k'm}\,\delta_{ij} - \frac{(p^+ + k^+)}{p^+}\,\delta_{jm}\,\delta_{ik'} - \frac{(p^+ + k^+)}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\\ &\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^{e}(x)\rho^{d}(y)\Big]\end{aligned}\tag{1.11}$$

Integrating w.r.t. w'

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_{\Lambda}^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ &\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\ &\Big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^{e}(x)\rho^{d}(y)\Big]\end{aligned}\tag{1.12}$$

Using + prescription we obtain the following

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_{0}^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ &\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\ &\Big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^{e}(x)\rho^{d}(y)\Big]\\ &-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{\vee - k^+}{\Lambda}\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ &\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,U^{cb}(z)\,\rho^{e}(x)\rho^{d}(y)\Big]\end{aligned}\tag{1.13}$$

Simplifying the Wilson lines we get

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = &\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_{0}^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ &\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\,\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\ &\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big]\\ &-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right) + \log\left(\frac{\vee}{k^+} - 1\right)\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\ &\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big]\end{aligned}\tag{1.14}$$

The BFKL term of this row, written out. It is inside line 1 of Eq. (1.14), in the $-\frac{1}{k^+}\,\delta_{im}\delta_{jk'}$ term: at large p^+ the phase $e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}$ lets only $y' \to z$ contribute to the y' integral, which gives $\frac{k^+}{p^+ + k^+}$ times the value at $y' = z$, and $\int_0^{\vee - k^+}\frac{dp^+}{p^+ + k^+} = \log\frac{\vee}{k^+}$

(Why_log_vee.pdf). Your equation stays as it is; it equals (B) + [your equation − (B)], with

$$\text{(B)} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\left[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\right]$$

$$\text{Eq. (1.14)} - \text{(B)} = \text{Eq. (1.14)} + \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\frac{1}{(p^++k^+)}$$
$$\times\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\left[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\right]$$

(B) $\propto \log\frac{\vee}{k^+}$ is the BFKL term (it is in BFKL_rows.pdf); Eq. (1.14) − (B) has no $\log\frac{\vee}{k^+}$: it is finite for $\vee\to\infty$. The $\log(\frac{\vee}{k^+}-1)$ of your line 2 is not a BFKL term: it cancels the subtraction in $\left[\frac{1}{p^+}\right]_+$ of line 1.

1.6 Row VI

[Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\int_{y'}\Big[\mathbb{A}^{\dagger\,(1)\,e'}_{k'}(p^++k^+,-y')U^{\,e'c'}(y')U^{d'b}(z)\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)$$
$$-\bar{\mathbb{A}}^{\dagger\,(1)\,c'}_{k'}(p^++k^+,-y')\bar{\mathbb{C}}^{\dagger\,d'ac'}_{1jik'}(p^++k^+,y'-w';p^+,w'-z)U^{d'b}(z)\Big]\times\Big[U^{bc}(z)\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\bar{\mathbb{A}}^{(1)\,c}_{j}(p^+,-z)$$
$$-\bar{\mathbb{A}}^{(1)\,a}_{i}(k^+,-w)\mathbb{A}^{(1)\,b}_{j}(p^+,-z)\Big]$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^{+2}}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',x',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(w'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\delta^{(2)}\left[w'-\frac{(p^++k^+)\,y'-p^+\,z}{k^+}\right]\times\left[\delta_{k'm}\,\delta_{ij}-\frac{(p^++k^+)}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{(p^++k^+)}{k^+}\,\delta_{im}\,\delta_{jk'}\right]$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]\quad(1.15)$$

Integrating w.r.t. w'

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\frac{1}{p^+}\,\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]\quad(1.16)$$

Using + prescription we obtain the following:

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^++k^+)}-\left[\frac{1}{p^+}\right]_+\,\delta_{jm}\,\delta_{ik'}-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\left[\log\left(\frac{k^+}{\Lambda}\right)+\log\left(\frac{\vee}{k^+}-1\right)\right]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]\quad(1.17)$$

The BFKL term of this row, written out. It is inside line 1 of Eq. (1.17), in the $-\frac{1}{k^+}\,\delta_{im}\delta_{jk'}$ term: at large p^+ the phase $e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}$ lets only $y'\to z$ contribute to the y' integral, which gives $\frac{k^+}{p^++k^+}$ times the value at $y'=z$, and $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\frac{\vee}{k^+}$ (Why_log_vee.pdf). Your equation stays as it is; it equals (B) + [your equation − (B)], with

$$\text{(B)} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]$$

$$\text{Eq. (1.17)} - \text{(B)} = \text{Eq. (1.17)} \; + \frac{1}{(2\pi)^3} \frac{ig^4 \, f^{c'd'a}}{4\pi^4} \frac{1}{k^+} \int_0^{\vee - k^+} \frac{dp^+}{2\pi} \frac{1}{(p^+ + k^+)}$$
$$\times \int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)} \frac{(y'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(y'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]$$
$$\times \left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\right]$$

(B) $\propto \log \frac{\vee}{k^+}$ is the BFKL term (it is in `BFKL_rows.pdf`); Eq. (1.17) – (B) has no $\log \frac{\vee}{k^+}$: it is finite for $\vee \to \infty$. The $\log(\frac{\vee}{k^+} - 1)$ of your line 2 is not a BFKL term: it cancels the subtraction in $\left[\frac{1}{p^+}\right]_+$ of line 1.